

Ex.no: 8	MINI PROJECT - I
	FINANCIAL DATA ANALYTICS

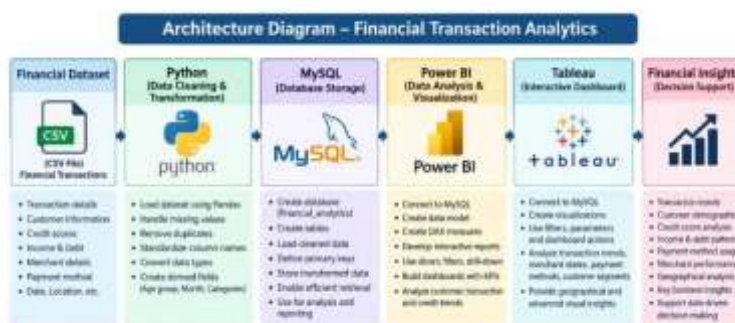
Aim:

To design, store, analyze, and visualize financial transaction data using SQL Server, Power BI, and Tableau to derive meaningful insights into revenue, cost, profit, profit margin, transaction volume, product categories, customer segments, regional performance, and financial trends.

Algorithm:

1. Create a financial analytics database in SQL Server.
2. Create relational tables for Regions, Customers, Products, and Transactions with appropriate primary and foreign keys.
3. Populate the database with financial master data and transaction records.
4. Prepare the transaction data containing transaction dates, customers, products, regions, quantities, revenue, and cost.
5. Connect the SQL Server database to Power BI for data modeling and analysis.
6. Clean and transform the data using Power Query by validating data types and creating derived fields such as Profit, Year, Quarter, and Month.
7. Create DAX measures for Total Revenue, Total Cost, Total Profit, Profit Margin, Total Transactions, and Total Quantity.
8. Develop interactive Power BI visualizations to analyze revenue trends, product categories, regional profitability, customer segments, and top-performing products.
9. Connect the same financial database to Tableau and establish relationships between the tables.
10. Create Tableau calculated fields for revenue, cost, profit, profit margin, and transaction count.
11. Develop Tableau worksheets and an interactive dashboard using KPIs, charts, and filters.
12. Analyze the dashboards to identify meaningful financial performance patterns.
13. End the process.

Architecture:



Program:

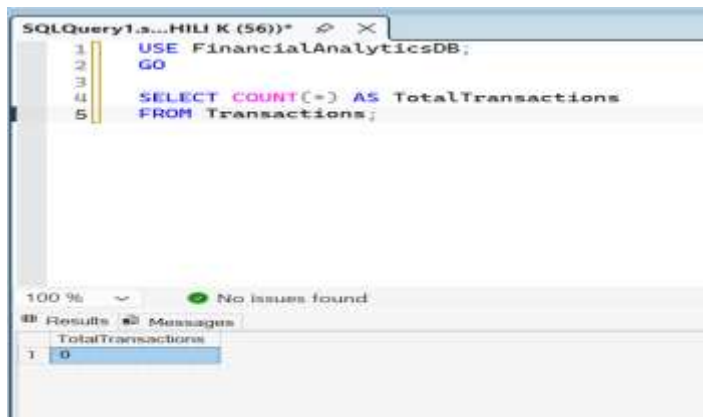
SQL Server – Database Creation

```
CREATE DATABASE FinancialAnalyticsDB;
```

```
GO
```

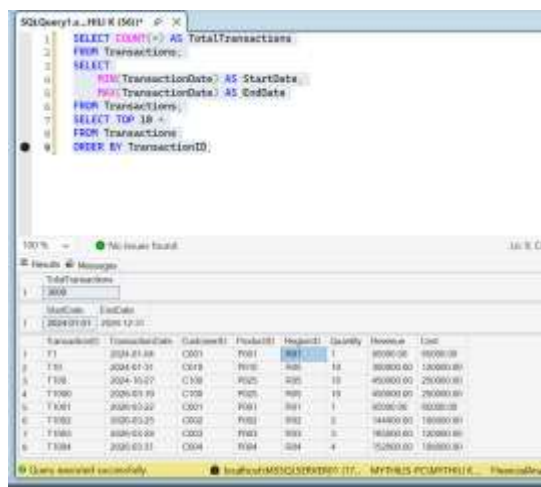
```
USE FinancialAnalyticsDB;
```

```
GO
```



SQL Server – Transaction Data

The database contains financial transaction information including:



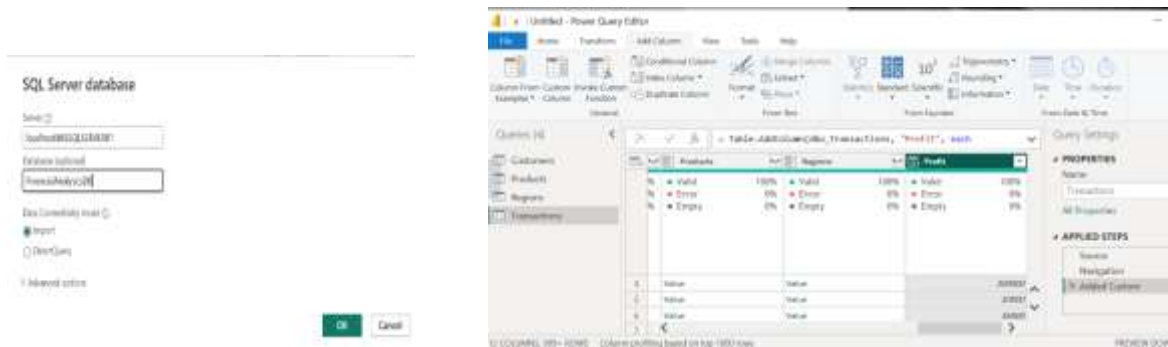
1- Power BI – Data Connection and Transformation

The FinancialAnalyticsDB database was connected to Power BI Desktop using the SQL Server connector.

The following tables were imported:

- Regions
- Customers
- Products
- Transactions

- A calculated **Profit** column was created using:
 $\text{Profit} = \text{Revenue} - \text{Cost}$



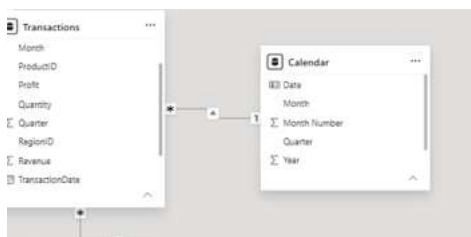
Power BI – Data Model

A relational analytical model was created in Power BI.



Power BI – Calendar Table

A Calendar table was created for time-based analysis.



. Power BI – DAX Measures

Total Revenue

Total Revenue = $\text{SUM}(\text{Transactions}[\text{Revenue}])$

Total Cost

Total Cost = $\text{SUM}(\text{Transactions}[\text{Cost}])$

Total Profit

Total Profit = $[\text{Total Revenue}] - [\text{Total Cost}]$

Profit Margin

Profit Margin % = DIVIDE([Total Profit], [Total Revenue], 0)

Total Transactions

Total Transactions =COUNTROWS(Transactions)

Total Quantity

Total Quantity =SUM(Transactions[Quantity])

Revenue Previous Year

Revenue Previous Year =

CALCULATE(
[Total Revenue],
SAMEPERIODLASTYEAR(Calendar[Date])
)

Revenue YoY Growth

Revenue YoY Growth % =DIVIDE([Total Revenue] - [Revenue Previous Year], [Revenue Previous Year], 0)

Profit Previous Year

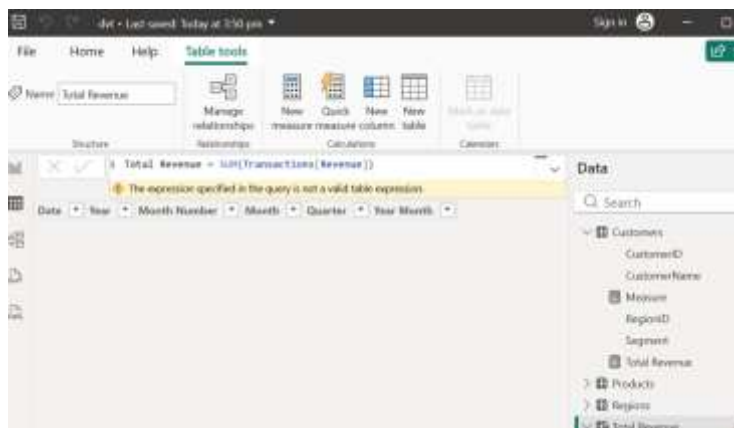
Profit Previous Year =

CALCULATE(
[Total Profit],SAMEPERIODLASTYEAR(Calendar[Date]))

Profit YoY Growth

Profit YoY Growth % =

DIVIDE([Total Profit] - [Profit Previous Year],
[Profit Previous Year], 0)



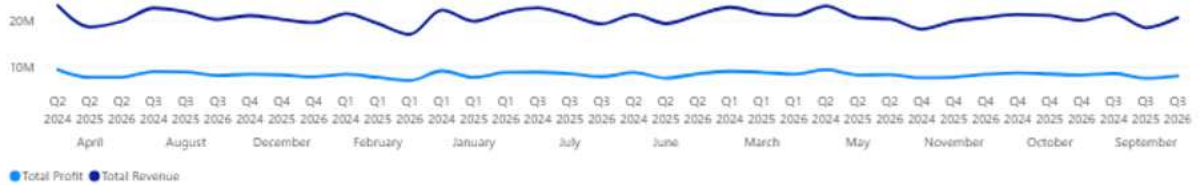
Data Visualization Using Power BI

KPI Cards: Card



Revenue & Profit Trend: Line Chart

Total Profit and Total Revenue
by Month, Year and Quarter



Revenue by Category: Column Chart , Profit by Region: Bar Chart



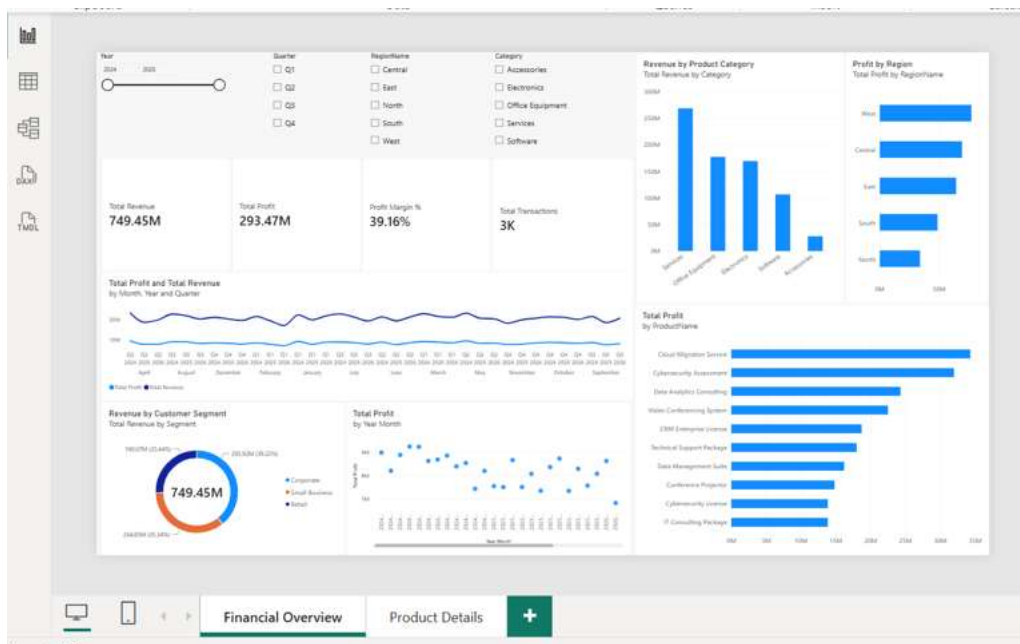
Revenue by Customer Segment: Donut Chart , Top 10 Products by Profit: Bar Chart



Interactive Slicers

Year	Quarter	RegionName	Category
2024 2026	☐ Q1	☐ Central	☐ Accessories
<input type="range"/>	☐ Q2	☐ East	☐ Electronics
	☐ Q3	☐ North	☐ Office Equipment
	☐ Q4	☐ South	☐ Services
		☐ West	☐ Software

Power BI Dashboard



Drill-down functionality was implemented in the trend analysis to navigate from Year → Quarter → Month.



2- Tableau – Data visualization

The same FinancialAnalyticsDB database was connected to Tableau using the SQL Server connector.



Tableau – Calculated Fields

Total Revenue - SUM([Revenue])

Total Cost - SUM([Cost])

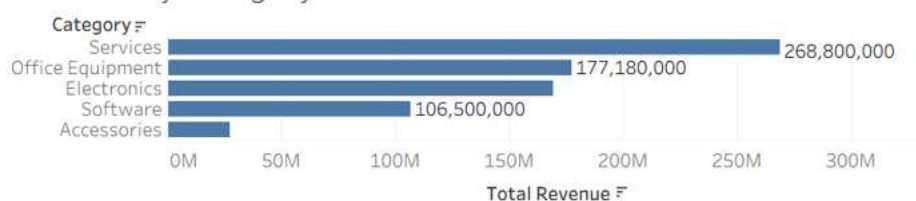
Total Profit - SUM([Revenue]) - SUM([Cost])

Profit Margin % - (SUM([Revenue]) - SUM([Cost])) / SUM([Revenue])

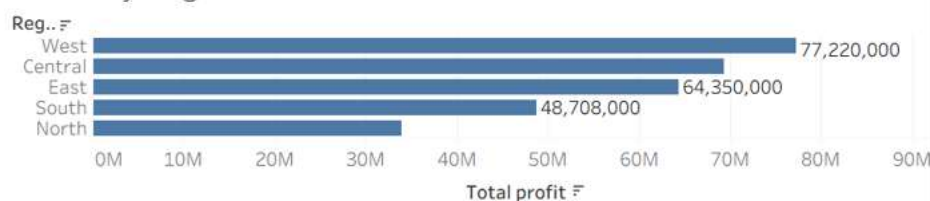
Total Transactions - COUNT([Transaction ID])

Tableau Dashboard

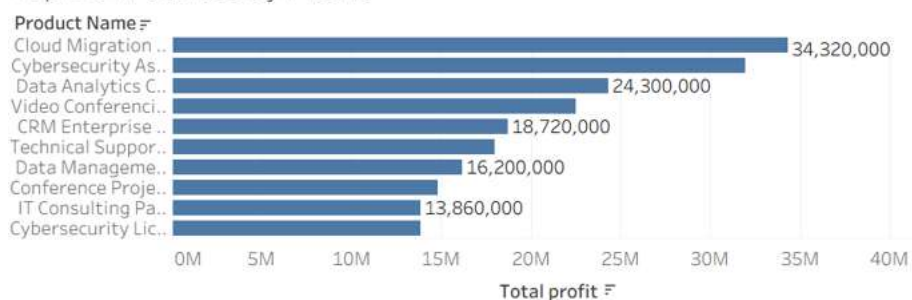
Revenue by Category



Profit by Region



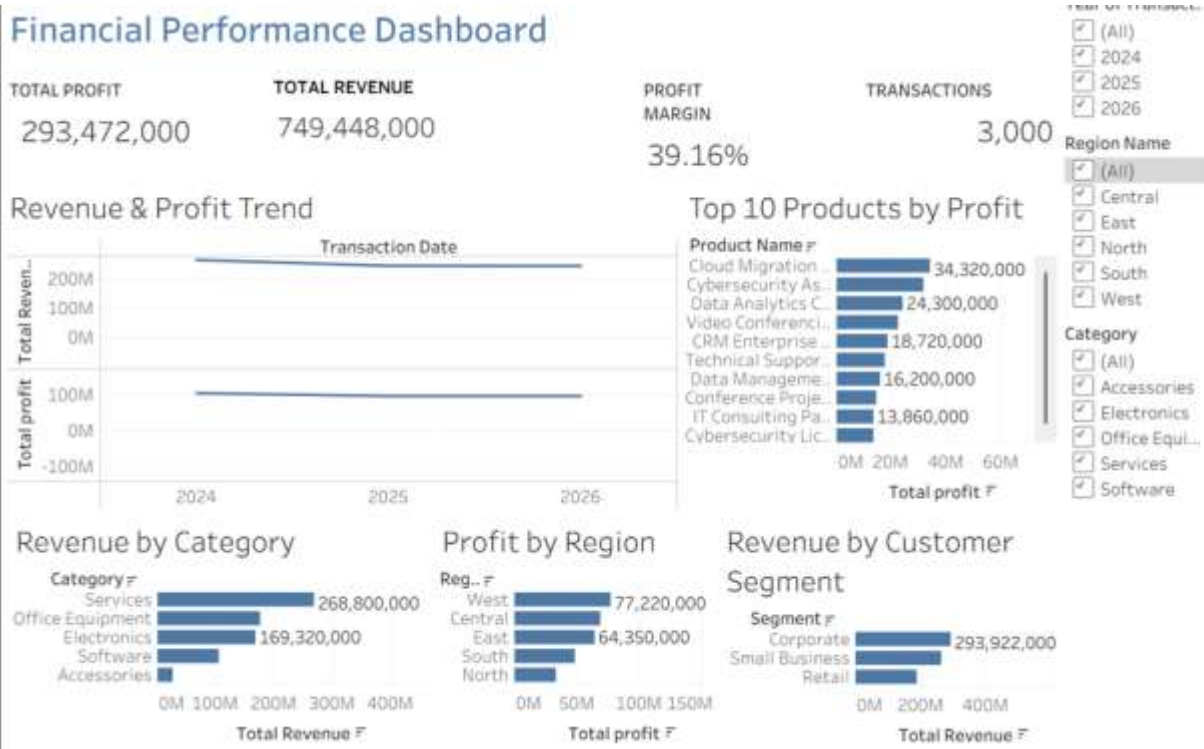
Top 10 Products by Profit



The dashboard includes:

- Revenue KPI
- Profit KPI
- Profit Margin KPI
- Transaction KPI
- Revenue and Profit Trend
- Revenue by Category
- Profit by Region

- Revenue by Customer Segment
- Top 10 Products by Profit



Result: